Unit 10 Lesson 9: Solving Problems Involving Tangents

Benchmark

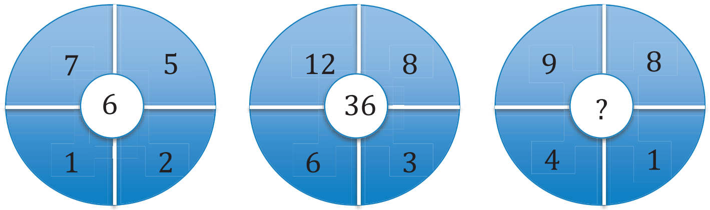
MA.912.GR.6.2 Solve mathematical and real-world problems involving the measures of arcs and related angles.

Learning Target

- [ ] I can solve problems related to tangents and secants and the measures of their intercepted arcs.

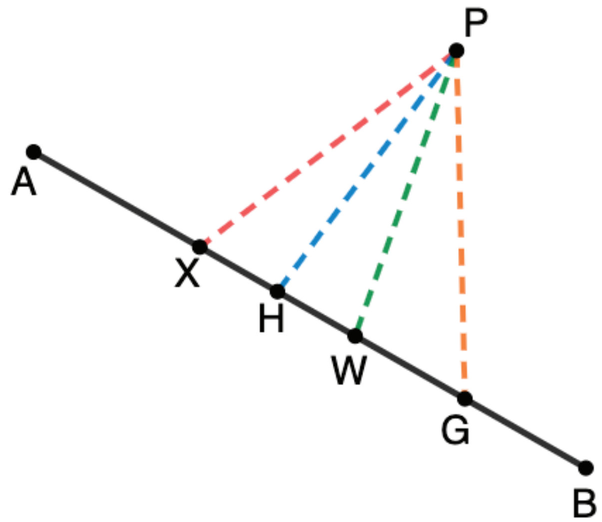
10.9.1 Warm-Up: Number Puzzle

1. In the circles, the outer numbers are used to calculate the number in the middle. Find the rule to determine what number should replace the question mark in the third circle.



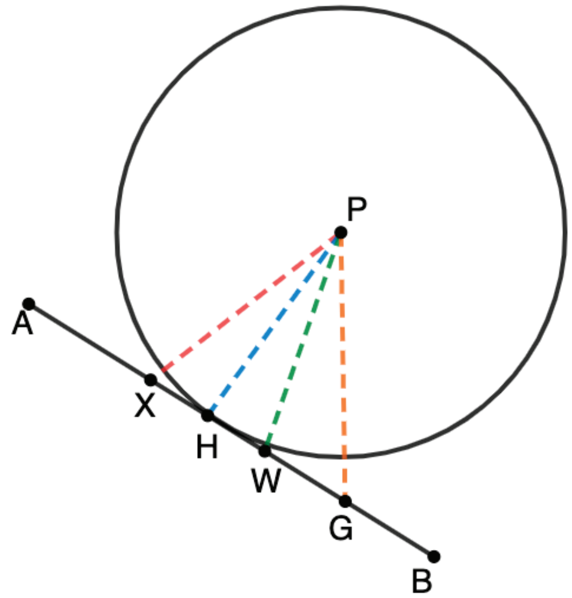
10.9.2 Exploration: Intersecting Tangent Lines

1. With your partner, answer the following.



a. Without measuring, which dashed line represents the shortest distance from point P to $\overline{AB}$? Explain your answer.

b. Point P is the center of a circle that intersects $\overline{AB}$. In the table, describe your observations of each dashed segment in relation to the center of Circle P.



Segment	**Relationship with the Center of Circle P**
$\overline{PG}$	
$\overline{PW}$	
$\overline{PH}$	
$\overline{PX}$	

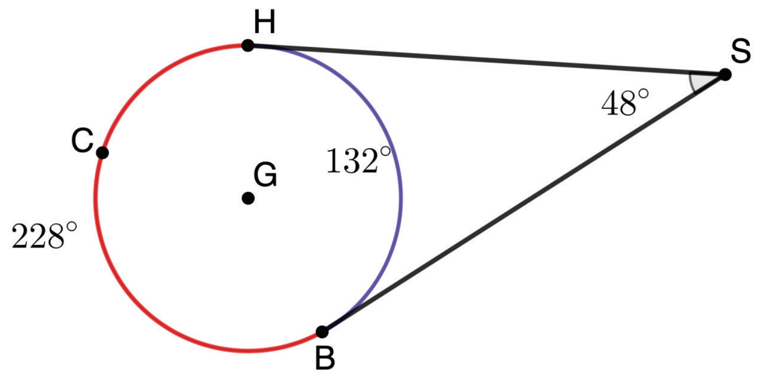
c. Discuss with your partner the relationship the blue segment has with the center of Circle P.

A ***tangent*** is a line or segment that intersects a circle at exactly one point.

If a line is tangent to a circle, then it is perpendicular to a radius, which ends at the point of tangency.

10.9.3 Collaboration: Angles Formed by Tangents and Secants

1. Circle G with tangents $\overline{SH}$ and $\overline{SB}$ is shown. Use the diagram to answer the following.



a. Complete the table.

	Measure
Major Arc HCB	
Minor Arc HB	
$\angle HSB$	

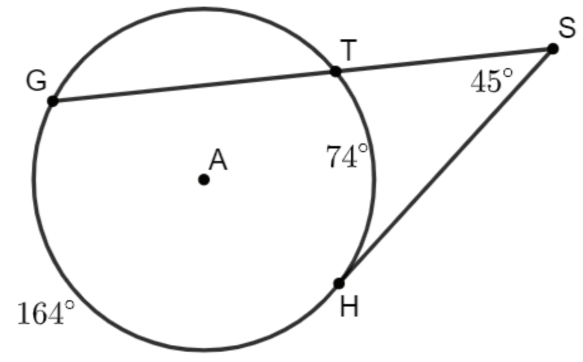
b. Use the values for the major and minor arcs of Circle G to determine a relationship between the arcs and the measure of the exterior angle formed by the two tangents. The relationship can be described using words or equations.

c. Ask two other partner pairs to share their relationships. Record their relationships and ask them to initial verifying you have written it correctly. If their description or equation is different than yours, discuss the differences and determine if your description or equation needs to be revised.

Description or Equation	Initials

d. Explain how to find the measure of an angle formed by two tangents drawn to a circle from a point outside that circle. Be sure to use the term(s) "intercepted arc(s)" at least once in the description.

2. Circle A shows $\angle GSH$ formed by the intersection of tangent $\overline{SH}$ and secant $\overline{GS}$ outside the circle at point S. $m\widehat{GH} = 164°$, $m\widehat{TH} = 74°$, and $m\angle GSH = 45°$.



a. Use the values for $\widehat{GH}$ and $\widehat{TH}$ of Circle A to determine the relationship between the arcs and the measure of the exterior angle formed by the tangent and the secant.

b. Discuss the relationship with your partner.

The formula for the measure of an **External Angle to a Circle** formed by two tangents, two secants, or a tangent and a secant is

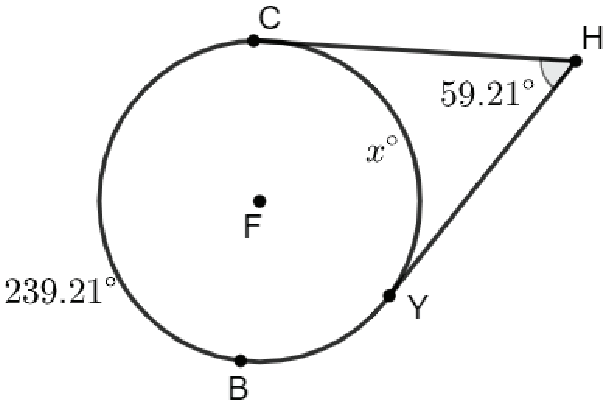
Measure of angle formed $= \frac{1}{2}(\text{difference of intercepted arcs})$

OR

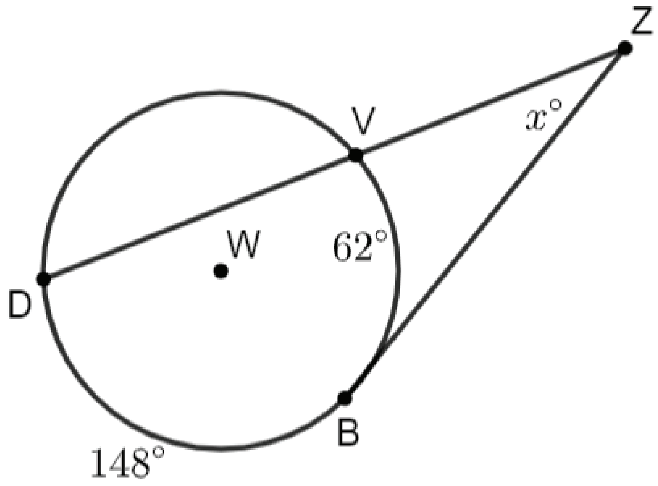
Measure of angle formed $= \frac{1}{2}(\text{larger arc} - \text{smaller arc})$

10.9.4 Your Turn!

1. Circle F has points C and Y on the circle. Tangents $\overline{CH}$ and $\overline{YH}$ intersect at point H outside the circle. $m\,\widehat{CBY} = 239.21°$, $m\,\widehat{CY} = x°$, and $m\angle CHY = 59.21°$. Find the value of x.



2. Circle W has points $D, V,$ and B on the circle. Tangent $\overline{BZ}$ and secant $\overline{DZ}$ intersect at Point Z outside the circle. $m\,\widehat{DB} = 148°$, $m\,\widehat{VB} = 62°$, and $m\angle DZB = x°$. What is $m\angle DZB$?



10.9.5 Wrap-Up: Reflection

Think back over this unit. If necessary, look back over the different lessons. Then, complete the sentences by checking all that apply.

1. I don't understand ...YET.
   - ☐ arc length
   - ☐ area of a sector
   - ☐ converting between radians and degrees
   - ☐ measures of arcs and related angles

2. I can explain
   - ☐ how to solve problems involving arc length.
   - ☐ how to solve problems involving: area of a sector.
   - ☐ converting angle measures between degrees and radians.
   - ☐ how to solve problems involving measures of arcs and related angles including central, inscribed, and intersections of a chords, secants, and tangents.

Unit 10, Lesson H1: Constructing Tangent Lines

Benchmark

MA.912.GR.5.5 Given a point outside a circle, construct a line tangent to the circle that passes through the given point.

Learning Target

☐ I can construct a line tangent to a circle that passes through a given point outside the circle.